Unless explicitly requested by a problem, do not include sketches as part of your proof. You are free to use the result from any problem on this (or previous) assignment as a part of your solution to a different problem even if you have not solved the former problem.

Problem 1 (2 pts)**.** Let S be a finite set of integers and let G be the graph on vertex set S where for any $x, y \in S$, $xy \in E(G)$ if and only if $x + y$ is odd.

1. Prove that G is a bipartite graph for any such S.

2. If $S = \{1, 2, \ldots, 100\}$, what is $|E(G)|$? Justify your answer.

Solution. 1. We can constrict a bipartition. Indeed, notice that $xy \in E(G)$ if and only if x and y have different parities. Consider thus $X = \{x \in S : x \text{ is even}\}$ and $Y = \{x \in S : x \text{ is odd}\}$, so that $V(G) = X \sqcup Y$. Since every edge of G connects a vertex in X to a vertex in Y, both X and Y are independent, so $X \sqcup Y$ is a bipartition of G.
2. As mentioned above, $E(G) = \{xy : x \in X, y \in Y\}$. Thus $|E(G)| = |X| \cdot |Y| = 50 \cdot 50 = 2500$. □

Problem 2 (2 pts)**.** Suppose that $(u = v_0, v_1, \ldots, v_k = v)$ is a u–v geodesic. Prove that $d(u, v_i) = i$ for all $i \in \{0, \ldots, k\}$.

Solution. Notice first for later use that $d(u, v) = k$, because $(u = v_0, v_1, \ldots, v_k = v)$.

Fix any $i \in \{0, \ldots, k\}$. Since$(u = v_0, v_1, \ldots, v_i)$ is a u–v_i of length i, we know that $d(u, v_i) \leqslant i$. Suppose for the sake of contradiction that $d(u, v_i) = \ell < i$; hence there is a u–v_i geodesic $(u = w_0, \ldots, w_\ell = v_i)$. But then $(u = w_0, \ldots, w_\ell = v_i, v_{i+1}, \ldots, v_k = v)$ is a u–v walk of length $\ell + (k - i)$. Theorem 1.6 then implies that there is a u–v path of length at most $\ell + (k - i) < k$, contradicting the previously established fact that $d(u, v) = k$. □

Problem 3 (2 pts)**.** Let G be a graph. For two non-empty subsets $A, B \subseteq V(G)$, an A–B path is a path in G which connects some vertex of A to some vertex of B. Prove that if P is a minim**al** A–B path, then P contains exactly one vertex from A and contains exactly one vertex from B.

Solution. Let $P = (v_0, \ldots, v_k)$ be a minimal A–B path with $v_0 in A$ and $v_k \in B$. In case $k = 0$ (which can happen only if A and B intersect), then we are trivially done, so we may assume that $k \geqslant 1$. Suppose for a contradiction that there is some $i \in \{1, \ldots k\}$ such that also v_i is in A (note that i could equal k if A and B intersect). But then $(v_0, \ldots, v_k)$ is also an A–B path which is a proper sub-path of P, a contradiction to the minimality of P. Therefore P contains exactly one vertex of A, namely v_0. A symmetric argument establishes that P contains exactly one vertex (namely v_k) of B. □

Problem 4 (2 pts)**.** Let G be a connected graph. Prove that any two maxim**um** paths in G must share some vertex.

Solution. Let k denote the length of any maximum path in G. Assume for a contradiction that $(v_0, \ldots, v_k)$ and $(u_0, \ldots, u_k)$ are two maximum paths in G with no vertices in common. Consider the subsets $A = \{v_0, \ldots, v_k\}$ and $B = \{u_0, \ldots, u_k\}$ of $V(G)$. Since G is connected, it contains an A–B path. Consider a minimal A–B path $P = (w_0, \ldots w_\ell)$ with $w_0 \in A$ and $w_\ell \in B$. Note that $\ell \geqslant 1$ since A and B are disjoint, and that problem 3 tells us all vertices of P other than w_0 and w_ℓ belong to neither A nor B. Since $w_0 \in A$ and $w_\ell \in B$, we have $w_0 = v_i$ and $w_\ell = u_j$ for some i, j. Consider the paths

$$P_1 = (v_0, \ldots, v_{i-1}, v_i = w_0, w_1, \ldots, w_{\ell-1}, w_\ell = u_j, u_{j+1}, \ldots, u_k),$$
$$P_2 = (v_k, \ldots, v_{i+1}, v_i = w_0, w_1, \ldots, w_{\ell-1}, w_\ell = u_j, u_{j-1}, \ldots, u_0).$$

Observe that the length of P_1 is $i + \ell + (k - j) \geqslant k + 1 + (i - j)$, where we used the fact that $\ell \geqslant 1$. Similarly, the length of P_2 is $(k - i) + \ell + j \geqslant k + 1 + (j - i)$ Since either $i \geqslant j$ or $j \geqslant i$, we see that one of P_1, P_2 has length at least $k + 1$. This contradicts the fact the length of a maximum path in G is k. □

Problem 5 (2 pts)**.** Prove that a graph G is bipartite if and only if every subgraph H of G has an independent set consisting of at least half of $V(H)$. (recall that an independent set is a set of vertices which induce no edges)

Solution. Assume first that G is bipartite and let $V(G) = X \sqcup Y$ be a bipartition of G. If H is any subgraph of G, we discussed in class that $(V(H) \cap X) \sqcup (V(H) \cap Y)$ is a bipartition of H. In particular both $V(H) \cap X$ and $V(H) \cap Y$ are independet sets of H, and since their union gives $V(H)$, at least one of them have size $\geqslant \frac{V(H)}{2}$.

Conversely, assume that G is not bipartite. Then by Theorem 1.12, G contains an odd cycle C, say of length n. Then C is a subgraph of G, and we showed in class that largest size of an independent set of C is $\frac{n-1}{2}$. □